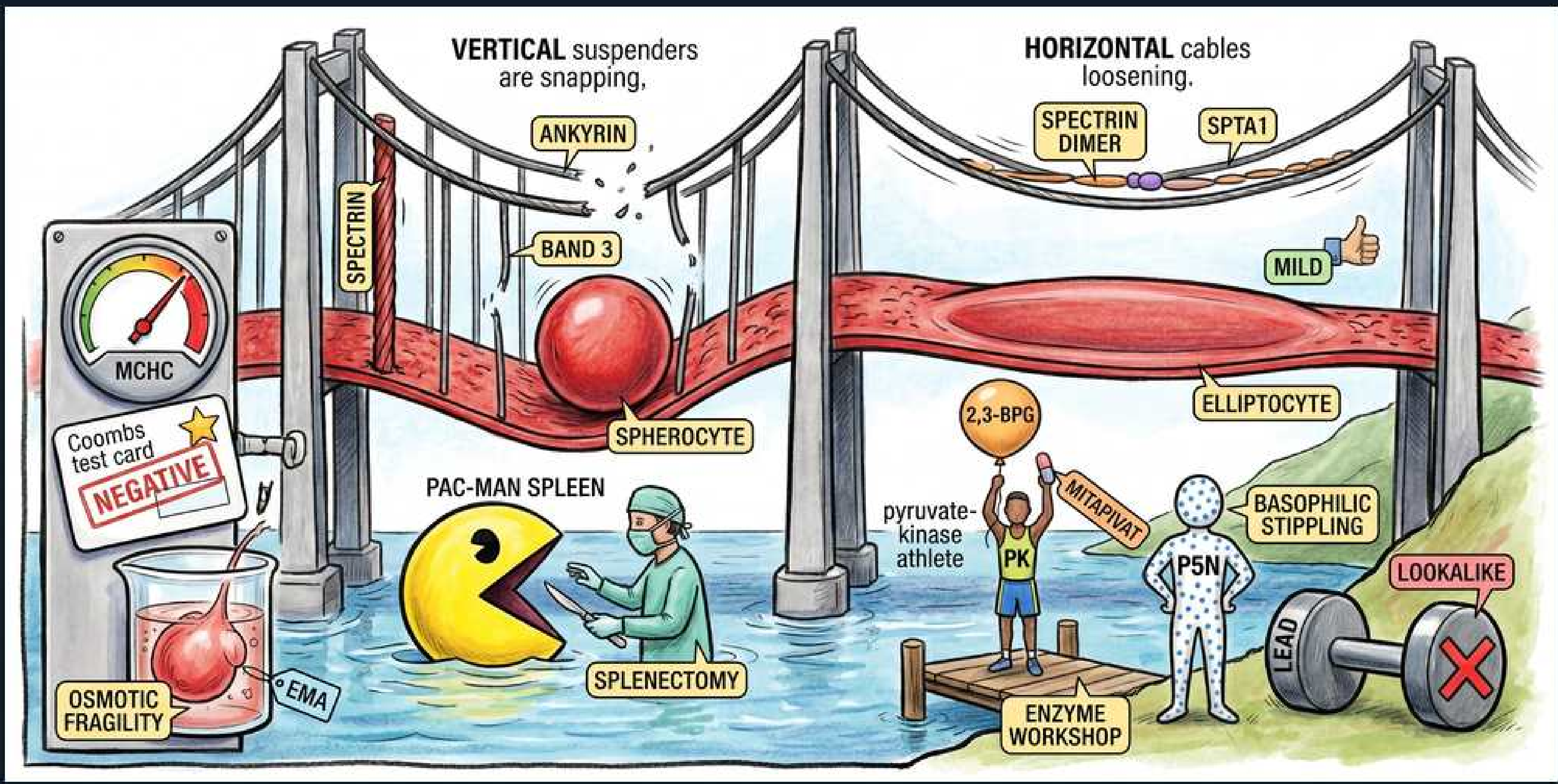
Inherited RBC Defects — Membrane (HS / HE) & Enzymes (PK, P5N)



1. Ankyrin — vertical defect

WHAT YOU SEE

The 'ANKYRIN' tag clipped to a VERTICAL suspender cable on the left span, beneath the 'VERTICAL suspenders are snapping' banner.

WHAT IT ENCODES

Hereditary spherocytosis is a defect in the VERTICAL membrane-skeleton links — ankyrin (most common), then spectrin, band 3, protein 4.2. Loss of vertical anchoring lets membrane bleb off so the cell rounds into a sphere.

BOARD TRAP

Calling ankyrin/HS a horizontal defect — vertical interactions = spherocytosis; horizontal = elliptocytosis.

2. Spectrin / band 3

WHAT YOU SEE

The vertical 'SPECTRIN' cable running down the left tower and the 'BAND 3' tag on the vertical suspenders.

WHAT IT ENCODES

Other HS genes are spectrin (SPTB), band 3 (SLC4A1) and protein 4.2; inheritance is autosomal dominant in about 75%.

BOARD TRAP

(mechanism)

3. Spherocyte

WHAT YOU SEE

A solid round red sphere with no central pallor hanging off the snapped left span, tagged 'SPHEROCYTE'.

WHAT IT ENCODES

Loss of surface-area-to-volume ratio yields a spherocyte with no central pallor — rigid, trapped and removed by the spleen (extravascular hemolysis).

BOARD TRAP

Spherocytes also occur in warm AIHA — the DAT distinguishes them (HS negative, AIHA positive).

4. MCHC high

WHAT YOU SEE

The wall-mounted 'MCHC' gauge with its needle pinned high into the red zone.

WHAT IT ENCODES

Hereditary spherocytosis is one of the few causes of a genuinely HIGH MCHC (cellular dehydration); the RDW is also elevated.

BOARD TRAP

Dismissing a high MCHC as artifact — in HS it is a real diagnostic clue.

5. Coombs NEGATIVE

WHAT YOU SEE

The 'Coombs test card' stamped 'NEGATIVE' below the MCHC meter.

WHAT IT ENCODES

HS is Coombs/DAT NEGATIVE — the key line separating it from warm AIHA, which is also spherocytic but DAT positive.

BOARD TRAP

Spherocytes + jaundice read as AIHA — check the DAT; if negative, think HS.

6. Osmotic fragility

WHAT YOU SEE

The 'OSMOTIC FRAGILITY' beaker of pink fluid with a cell lysing inside it.

WHAT IT ENCODES

HS cells lyse readily in hypotonic saline → increased osmotic fragility (the classic test).

BOARD TRAP

A normal osmotic fragility read as excluding HS — it can miss mild cases; the modern test of choice is EMA binding.

7. EMA binding

WHAT YOU SEE

The 'EMA' label tagged beside the osmotic-fragility beaker on the lab panel.

WHAT IT ENCODES

Eosin-5-maleimide (EMA) binding measured by flow cytometry is reduced in HS and is now the preferred diagnostic test.

BOARD TRAP

(best test)

8. Spleen / splenectomy

WHAT YOU SEE

The yellow PAC-MAN SPLEEN chomping cells in the water, tagged 'SPLENECTOMY' with the masked surgeon beside it.

WHAT IT ENCODES

The spleen is the site of destruction; splenectomy cures the anemia (cells survive) but is reserved for moderate-severe disease, deferred past age 5–6, with pre-op vaccination against encapsulated organisms and lifelong sepsis/thrombosis risk.

BOARD TRAP

Splenectomy for everyone — it is reserved for significant disease, and gallstones still form.

9. Spectrin dimer / SPTA1 — horizontal defect

WHAT YOU SEE
The 'SPECTRIN DIMER' and 'SPTA1' tags on the HORIZONTAL cables loosening across the right span, under 'HORIZONTAL cables loosening'.

WHAT IT ENCODES
Hereditary elliptocytosis is a defect in HORIZONTAL spectrin–spectrin (dimer–dimer) interactions; SPTA1 (alpha-spectrin) is the most common gene.

BOARD TRAP
Calling a horizontal defect a cause of spheres — horizontal = ellipses.

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10. Elliptocyte — usually mild

WHAT YOU SEE
A stretched oval red cell slung along the loosened horizontal right span, tagged 'ELLIPTOCYTE' beside a green thumbs-up 'MILD'.

WHAT IT ENCODES
Cells form elongated ellipses; most hereditary elliptocytosis is clinically mild or asymptomatic and needs no treatment.

BOARD TRAP
Assuming HE always hemolyzes — it is usually mild; hereditary pyropoikilocytosis (HPP) is the severe variant.

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11. Pyruvate kinase (PK)

WHAT YOU SEE
The blue 'PK' pyruvate-kinase athlete standing on the enzyme-workshop dock below the spans.

WHAT IT ENCODES
Pyruvate kinase deficiency is the most common glycolytic enzyme defect — AUTOSOMAL RECESSIVE — causing chronic nonspherocytic hemolytic anemia.

BOARD TRAP
Calling PK X-linked (that is G6PD) — PK deficiency is autosomal recessive.

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12. 2,3-BPG

WHAT YOU SEE
The PK athlete hoisting a yellow barbell labeled '2,3-BPG' overhead on the dock.

WHAT IT ENCODES
Because PK acts downstream, the block makes 2,3-BPG accumulate, right-shifting the oxygen-dissociation curve and improving tissue O_2 delivery, so the anemia is relatively well tolerated for the hemoglobin level.

BOARD TRAP
(explains good tolerance)

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13. Mitapivat

WHAT YOU SEE
The 'MITAPIVAT' tag pinned across the PK athlete as a performance booster.

WHAT IT ENCODES
Mitapivat, an oral PK activator, is the first approved disease-specific drug for PK deficiency; splenectomy also helps.

BOARD TRAP
(treatment)

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14. P5N — basophilic stippling

WHAT YOU SEE
The grey 'P5N' athlete dotted with coarse blue speckles, tagged 'BASOPHILIC STIPPLING'.

WHAT IT ENCODES
Pyrimidine-5'-nucleotidase (P5N) deficiency causes coarse basophilic stippling from undegraded ribosomal RNA.

BOARD TRAP
(see lead trap)

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15. Lead — lookalike

WHAT YOU SEE
The grey 'LEAD' dumbbell with a red X, tagged 'LOOKALIKE', planted next to the P5N athlete.

WHAT IT ENCODES
LEAD poisoning inhibits P5N and phenocopies it, producing identical coarse basophilic stippling.

BOARD TRAP
Calling basophilic stippling always inherited P5N — lead poisoning is the acquired lookalike; check a lead level.
